AI 500
Foundations of Artificial Intelligence



L9: Causal Inference

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LUMS
A Not-for-Profit University

Causal Questions

Causal questions

Education / Social Science Examples



- Does smaller class size cause better student performance?
- Does online learning cause lower engagement compared to in-person teaching?
- Does teacher feedback frequency cause higher exam scores?
- What is the causal effect of scholarship offers on college enrollment?
- Does peer group composition affect student achievement?



Causal questions

Health and Medicine

- Does smoking cause lung cancer?
- What is the effect of a new drug on blood pressure?
- Does air pollution increase the incidence of asthma?
- Does exercise reduce the risk of depression?
- Does vitamin D supplementation cause stronger bone density in children?



Causal questions

Economics and Policy



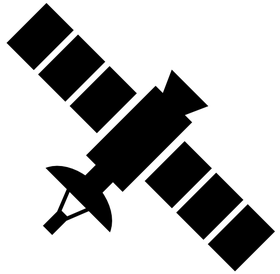
- Does increasing the minimum wage cause higher unemployment?
- Does universal basic income improve mental health outcomes?
- Does tax incentives for electric vehicles increase EV adoption?
- What is the effect of remote work policies on productivity?



Causal questions

Urban Planning / Remote Sensing

- Does green space availability reduce urban heat islands?
- Does proximity to public transport cause higher property values?
- Does zoning regulation influence building density?
- What is the causal effect of urban expansion on flood risk?





Causality and Data

Data Generating Process



- To draw meaningful insights from data, we need to understand how the data were generated
- Were the data generated from
  - a collection in nature (e.g., changes in pollution levels in Lahore)
  - an industrial process (e.g., heat generated during manufacturing)
  - an experiment (e.g., evaluation of a new drug on mice)
- Do we have census data or a sample?
  - was a chance mechanism used to select the sample?
  - or, is this a self-selected or convenience sample?

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Causality and Data

Observation is a key to good science



- An **observational** study is one in which scientists make conclusions based on data that they have **observed** but had no hand in generating
- In data science, many such studies involve observations on a group of individuals
  - a factor of interest called a **treatment**, and an **outcome** measured on each individual

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Individuals, Treatments, & Outcomes

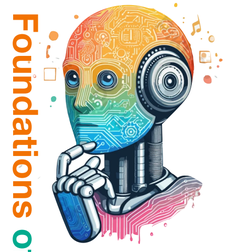


- In a study of whether chocolate is good for the health
  - individuals would be people
  - treatment would be eating chocolate and
  - outcome might be a measure of heart disease

- In a study of whether the capital punishment has any deterrent effect
  - the individuals could be the states in the US
  - a state law allowing the death penalty would be the treatment
  - and an outcome could be the state's murder rate

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Fundamental Question



> *"Does the **treatment** have an effect on the **outcome**?"*

- Any relation between the **treatment** (X) and the **outcome** (Y) is called an association
- If the **treatment** (X) causes the **outcome** (Y) to occur, then the association is causal

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Association vs. Causation

Good predictions need not have good explanations.

The owl can be a good hunter without understanding why the rat always goes from point A to point B

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Association (Correlation)



- Two variables are associated if they tend to vary together, when one changes, the other also tends to change, but not necessarily because one causes the other.

Example

- Ice cream sales ↑ and drowning incidents ↑.

- → They are associated because both rise in summer.
  → But ice cream does not cause drowning.

Key idea

- Association is about statistical dependency, not mechanism.

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Association (Correlation)

Mathematical View



You can detect association using:

- Correlation coefficients $\rho(X, Y)$
- Mutual information $I(X; Y)$
- Regression coefficients (nonzero slope)

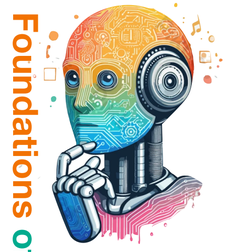
But all of these are symmetric:

- $\rho(X, Y) = \rho(Y, X)$

- So they don't tell us which causes which.

Causation

Causation requires intervention



Definition

- Given, **treatment** (X) and **outcome** (Y)
- X causes Y if changing X (while holding everything else constant) changes Y.
- Formally, in the potential outcomes framework:

$$X \text{ causes } Y \quad \text{if} \quad E[Y(1)] \neq E[Y(0)]$$

- That is, the expected value of Y differs under two different states of X.

Key to Establishing Causality



- If the treatment and control groups are similar apart from the treatment, then differences between the outcomes in the two groups can be ascribed to the treatment.

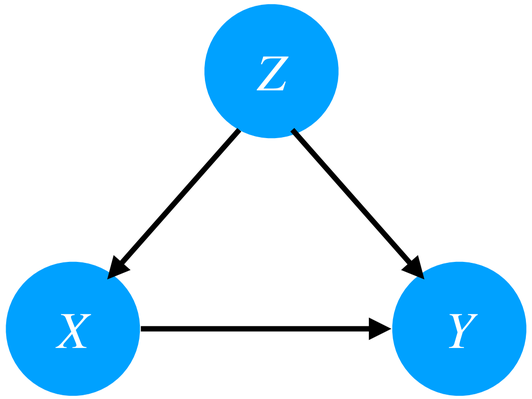
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Confounding and Bias

Association ≠ Causation

WHY?

- A confounder Z is a third variable that affects both X and Y, creating a spurious association.
- **Example**:
  - $Z =$ temperature
  - $X =$ ice cream sales
  - $Y =$ drowning incidents
- Here, X and Y are associated through Z, but X does not cause Y



Example: Coffee and Lung Cancer



- Studies in the 1960's showed that **coffee drinkers** had **higher rates of lung cancer** than those who did not drink coffee
- Because of this, some people identified coffee as a **cause of lung cancer**
  - But coffee does not cause lung cancer!
- What was happening?

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Example: Coffee and Lung Cancer



- The analysis contained a **confounding factor**—smoking
- In those days, coffee drinkers **were also likely to have been smokers**, and smoking does cause lung cancer
- Coffee drinking was associated with lung cancer, but it did not cause the disease
- Confounding factors are **common in observational studies**
  - Good studies take great care to reduce confounding and its effects

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London (Miasmas and Miasmatists)

Causes of Cholera outbreak in London (1832, 1849, 1854, etc.)

Historical Context: London in the 1800s

Miasma and Anti-Miasmatists Theories



- In the early to mid-1800s, London was overcrowded, polluted, and poorly sanitized.
- There was lot of air pollution.
- There was **no proper sewage system**.
- Human waste and industrial runoff flowed into the **River Thames**, which was also the **city's drinking water source**.
- Recurrent **cholera epidemics** (1832, 1849, 1854, etc.) killed tens of thousands.
- At that time, two competing theories tried to explain how disease spread The Miasma Theory and Anti-Miasmatists

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The Miasma Theory

“Miasma” comes from the Greek word for “pollution” or “bad air.”



- Miasmatists (those who supported the theory) believed that:
  - Diseases like cholera, typhus, and plague were caused by “**noxious vapors**” from decaying organic matter.
  - These vapors (miasmas) rose from **filthy streets, open sewers, and graveyards.**
  - Therefore, cleaning the **air**—through better ventilation and odor control—was seen as the best way to fight disease.
  - To protect themselves against infection, those who could afford would hold sweet-smelling things to their noses

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John Snow & Cholera

John Snow (1813–1858)



- For several years, he had been following the devastating waves of cholera that hit England from time to time
- The disease arrived suddenly and was almost immediately deadly
- He had noticed that while entire households were wiped out by cholera, the people in neighboring houses sometimes remained completely unaffected

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So, what was happening?

Snow’s Observation



- Snow had noticed that the onset of the disease almost always involved vomiting and diarrhea
- He therefore believed that the infection was carried by something people ate or drank, not by the air that they breathed
- His prime suspect was water contaminated by sewage

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August of 1854



- At the end of August 1854, cholera struck in the overcrowded Soho district of London
- As the deaths mounted, Snow recorded them diligently, using a method that went on to become standard in the study of how diseases spread: he drew a map
- On a street map of the district, he recorded the location of each death

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Snow’s Original Map



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Final Discovery



- Later, it was discovered that a cesspit that was just a few feet away from the well of the Broad Street pump had been leaking into the well
- Thus, the pump's water was contaminated by sewage from the houses of cholera victims!
- Snow used his map to convince local authorities to remove the handle of the Broad Street pump
- Though the cholera epidemic was already on the wane when he did so, it is possible that the disabling of the pump prevented many deaths

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Stuff of Legend?



- The removal of the Broad Street pump handle has become the stuff of legend
- The map gave Snow a strong indication that the cleanliness of the water supply was the key to controlling cholera
- Yet, he was still a long way from a convincing scientific argument that contaminated water was causing the spread of the disease
- To make a more compelling case, he had to use the method of comparison

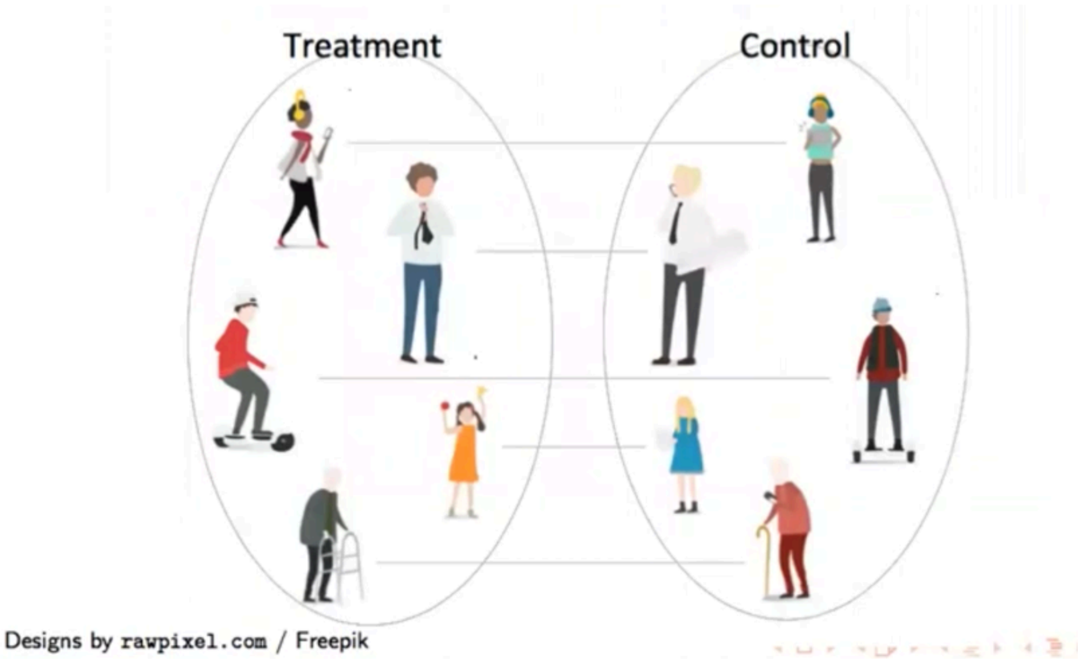
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Comparison



- Scientists use comparison to identify an association between a **treatment** and an **outcome**

- They compare the **outcomes** of a group of individuals who got the **treatment** (the **treatment** group) to the **outcomes** of a group who did not (the control group)

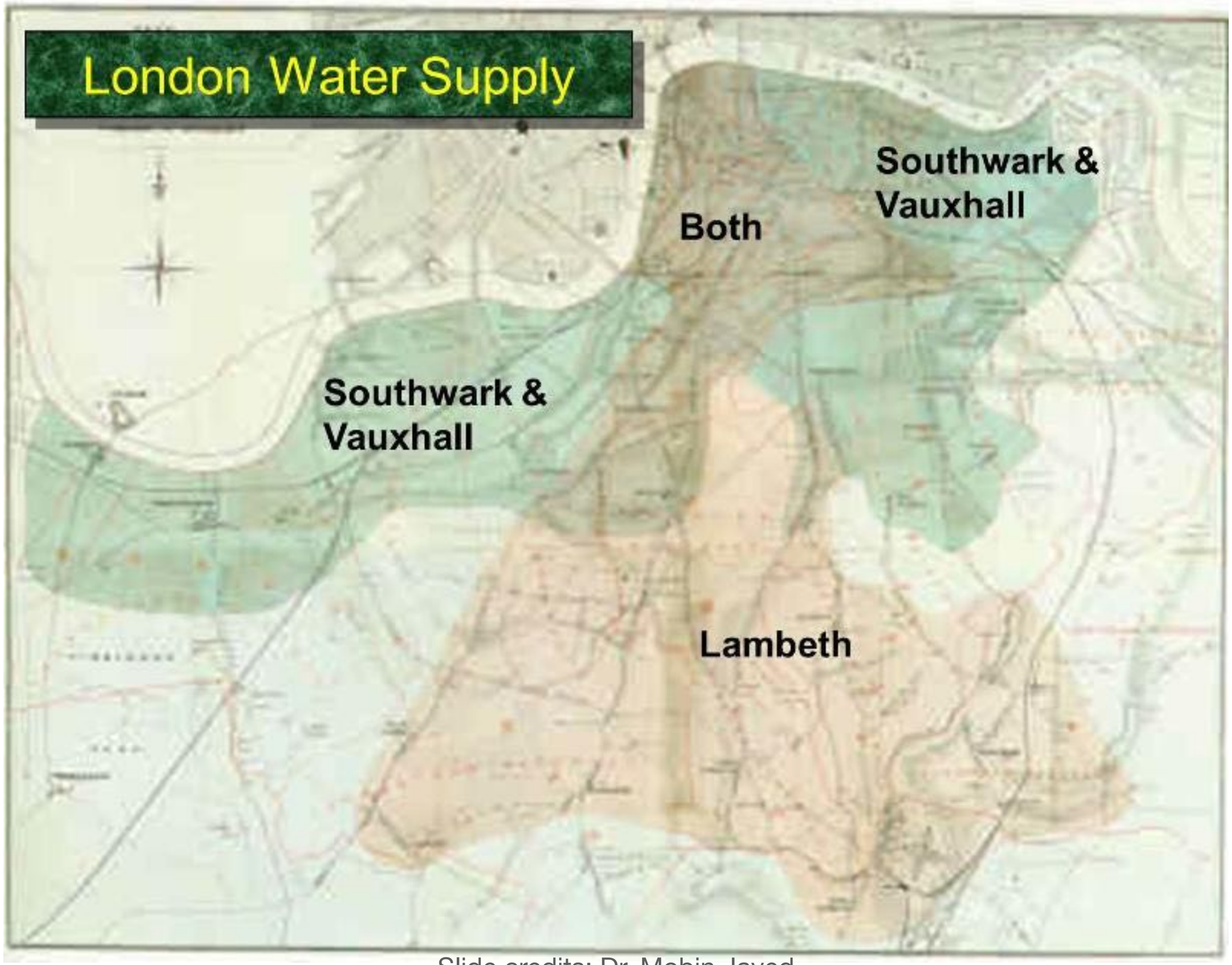


Snow's "Grand Experiment"



- Encouraged by what he had learned in Soho, Snow completed a more thorough analysis
- He had been gathering data on cholera deaths in an area of London that was served by two water companies

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No Systematic Difference



- Snow noticed that there was no systematic difference between the people who were supplied by S&V and those supplied by Lambeth
  - “Each company supplies both rich and poor, both large houses and small; there is no difference either in the condition or occupation of the persons receiving the water of the different Companies …”

Snow's "Grand Experiment"

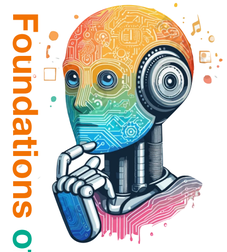


"…there is no difference whatsoever in the houses or the people receiving the supply of the two Water Companies, or in any of the physical conditions with which they are surrounded…"

- The two groups were similar except for the treatment

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Snow's Table



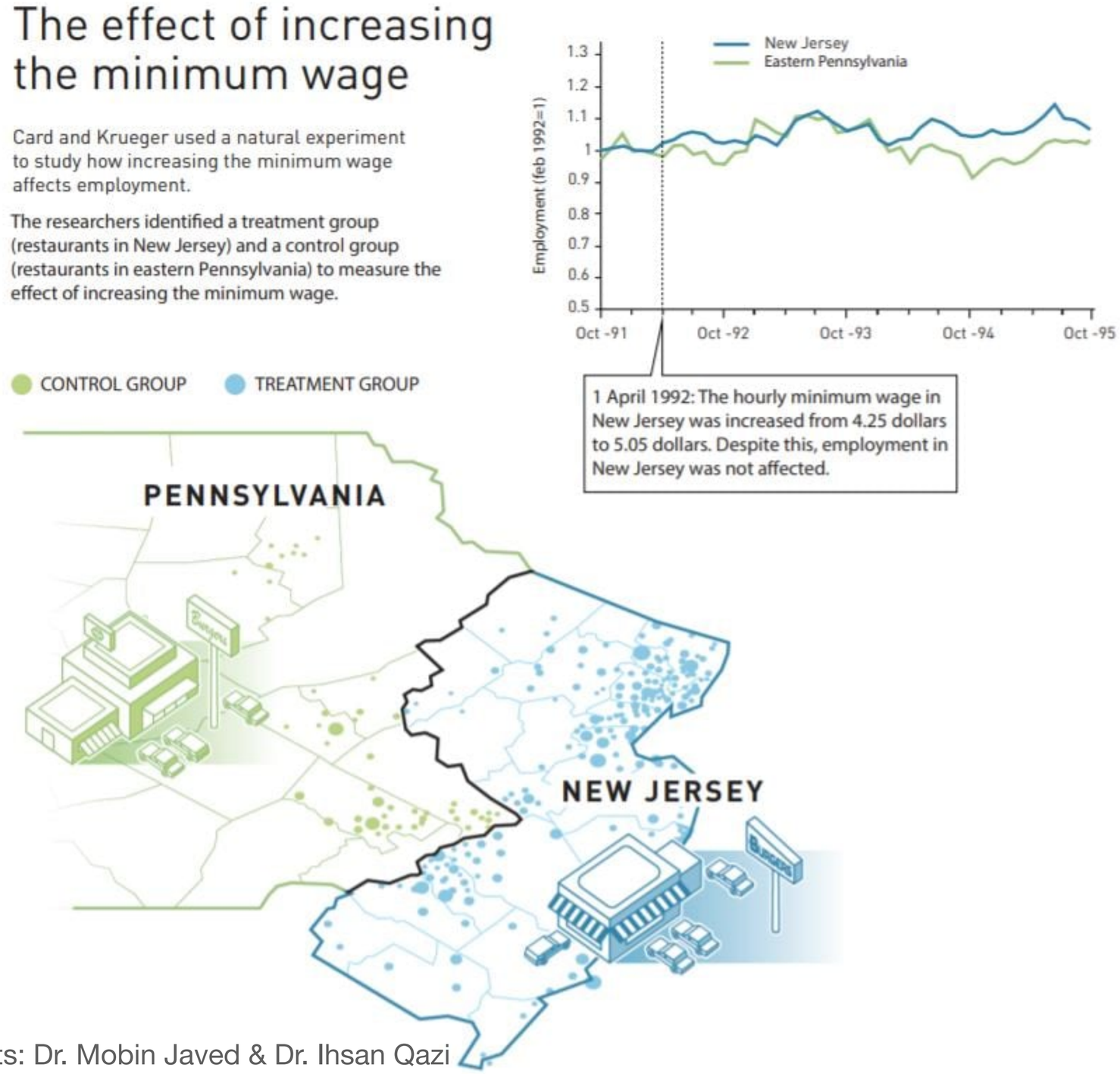
Supply Area	Number of houses	Cholera deaths	Deaths per 10,000 houses
S&V	40,046	1,263	315
Lambeth	26,107	98	37
Rest of London	256,423	1,422	59

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Understanding labour markets

- The effects of a minimum wage

- In the early 1990s, the conventional wisdom among economists was that higher minimum wages lead to lower employment because they increase wage costs for businesses.



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Counterfactual

Counterfactual



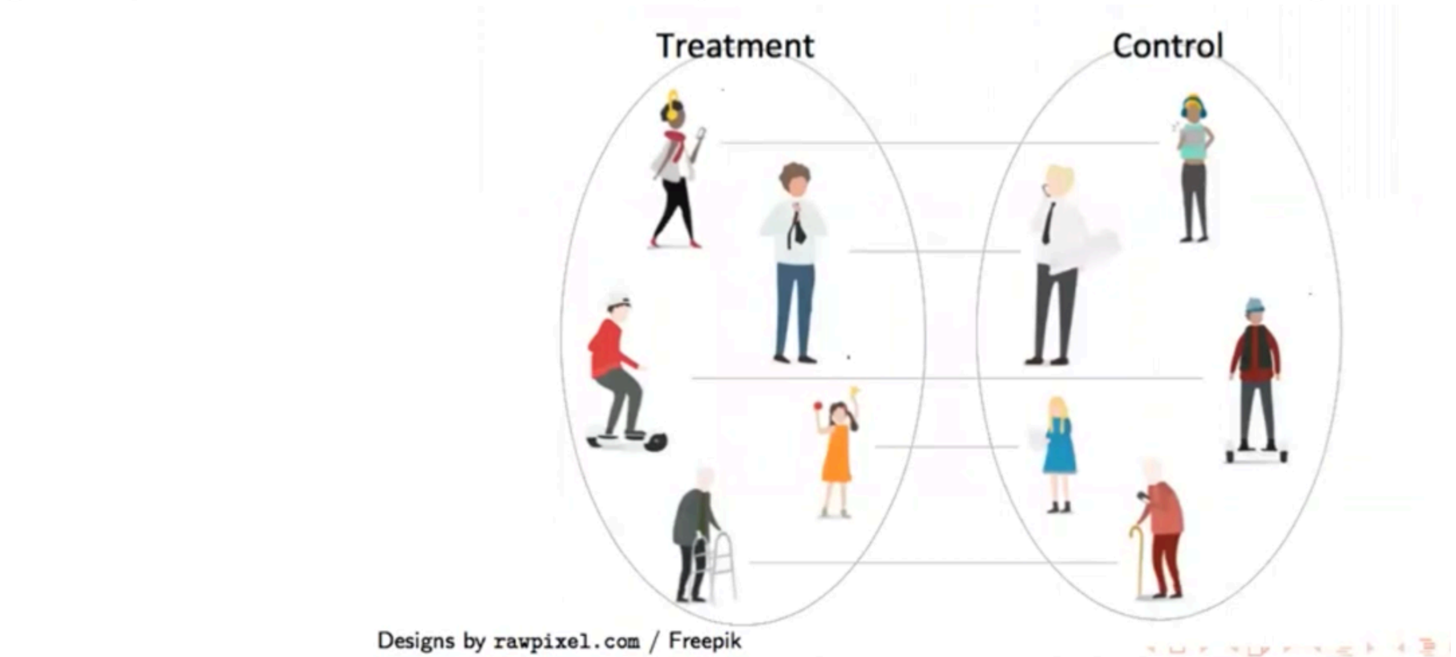
- The counterfactual represents the state of the world that program participants would have experienced in the absence of the program
- Problem: Counterfactual cannot be observed
- Solution: We need to "mimic" or construct the counterfactual

What if the patient had not taken the cholesterol lowering drug? What if an algorithm was not changed?

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Constructing the Counterfactual

- Usually done by selecting a group of individuals that did not participate in the program
  - This group is usually referred to as the control group or comparison group
- How this group is selected is a key decision in the design of any impact evaluation



Snow’s Brilliance



- Snow’s brilliance lay in identifying two groups that would make his comparison clear
- Snow’s methods led to the development of the field of epidemiology, which is the study of the spread of diseases

Correlation doesn't imply causation

How do we establish causality then?

Why are we interested in causality?

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"Some tens of thousands of years ago, humans began to realize that **certain things cause other things and that tinkering with the former can change the latter**. No other species grasps this, certainly not to the extent that we do. From this discovery came organized societies, then towns and cities, and eventually the science- and technology-based civilization we enjoy today. All because we asked a simple question: Why?"

The Book of Why – Judea Pearl and Dana Mackenzie

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Correlation vs. Causation

Urban Planning / Remote Sensing



- Does green space availability reduce urban heat islands?
- Does proximity to public transport cause higher property values?
- Does zoning regulation influence building density?
- What is the causal effect of urban expansion on flood risk?

Potential Outcome Framework

Neyman 1923, Rubin 1974

Recap: Causal questions

Education Examples

- Does smaller class size cause better student performance?



Health and Medicine

- Does drinking coffee causes lung cancer?



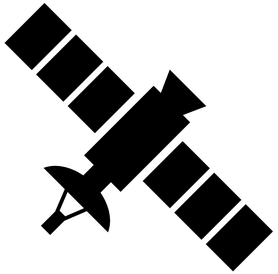
Economics and Policy

- Does tax incentives for electric vehicles increase EV adoption?



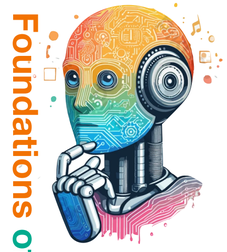
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Fundamental Question



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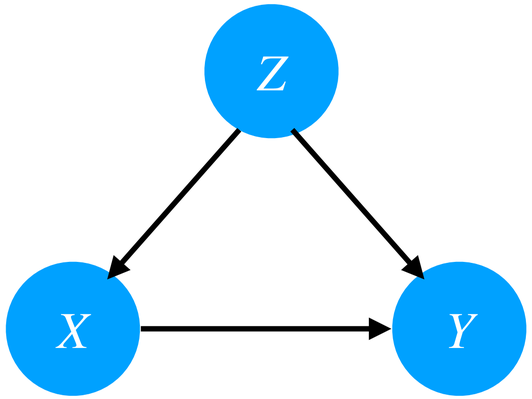
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WHY?

- A confounder Z is a third variable that affects both X and Y, creating a spurious association.
- **Example**:
  - $Z =$ temperature
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Prediction vs. Causation



Individuals, Treatments, & Outcomes



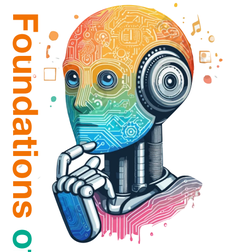
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Causation

Causation requires intervention



Definition

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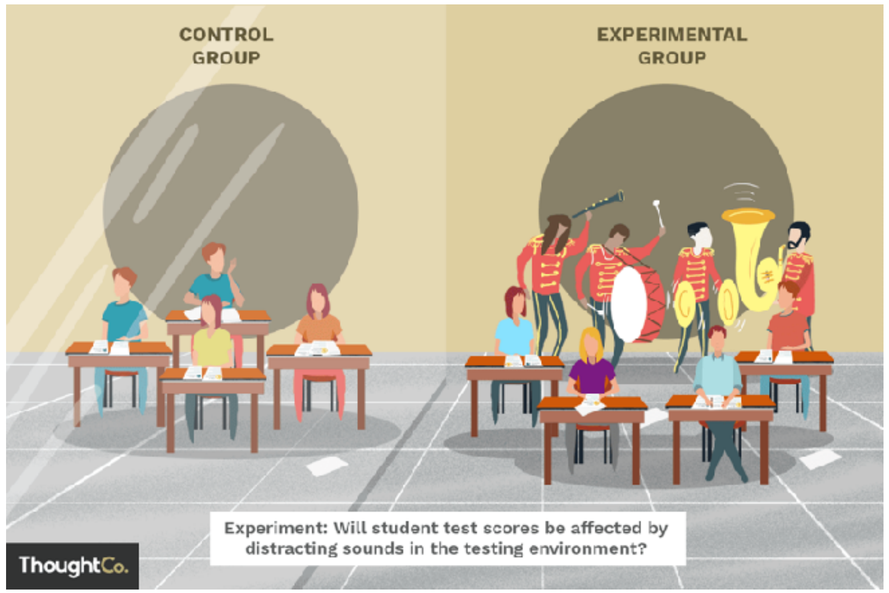
$$X \text{ causes } Y \quad \text{if} \quad E[Y(1)] \neq E[Y(0)]$$

- That is, the expected value of Y differs under two different states of X.

Potential Outcomes

Neyman 1923, Rubin 1974

- Let $Y_i(1)$ be the potential outcome of person i under treatment and $Y_i(0)$ be his potential outcome under control
- For e.g. example,
  - $Y_i(1)$ = income that the ith person woulh have if he praticipates in the program
  - $Y_i(0)$ = income that the ith person woulh have if he **does not** praticipates in the program



$Y_i(0)$ $Y_i(1)$

- $Y_i(1)$ is observed only for the people that receive the treatment, otherwise it is unobserved or counterfactual

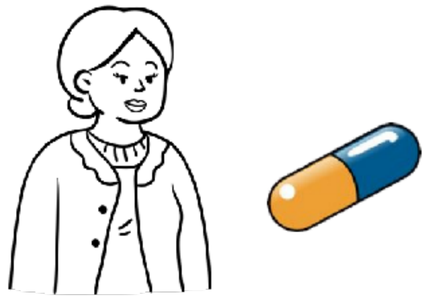
Treatments, Controls, and Outcomes



- We will assess causality with respect to a binary variable T, and refer to its values as the treatment and the control
  - We use variable $i \in \{1, \ldots, n\}$ to denote the experimental units
- To each unit, we apply either the treatment or the control, but not both
  - $\{T_i = 1\}$: denotes the event that unit i is given the treatment and
  - $\{T_i = 0\}$: denotes the event that unit i is given the control
- The observed outcome for unit i is denoted by Y_i

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Potential Outcomes & Counterfactuals



$Y_i(1)$



$Y_i(0)$

$$Y_i = \begin{cases} Y_i(1) \; if \; T_i = 1 \\ Y_i(0) \; if \; T_i = 0 \end{cases}$$

Observed outcome for unit i



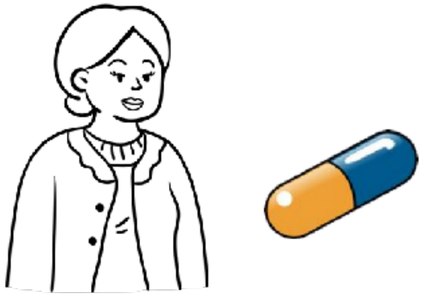
Potential outcome if person $_i$ received treatment



Potential outcome if person $_i$ did not receive treatment

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Potential Outcomes & Counterfactuals



$$Y_i(1)$$



$$Y_i(0)$$

Individual causal effect of treatment on unit $_i$:

$$\tau_i = Y_i(1) - Y_i(0)$$



Potential outcome if person $_i$ received treatment



Potential outcome if person $_i$ did not receive treatment

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Fundamental Problem of Causal Inference

- We cannot observe the counterfactual!
  - i.e., we can only observe one of the potential outcomes!

Person (i)	T_i	$Y_i(1)$	$Y_i(0)$
1	1	0.6	0.3
2	0	0.2	0.6
3	1	0.4	0.2
4	0	0.3	0.1
5	1	0.7	0.5
6	0	0.6	0.4
7	0	0.3	0.1

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7	0	0.3	0.1

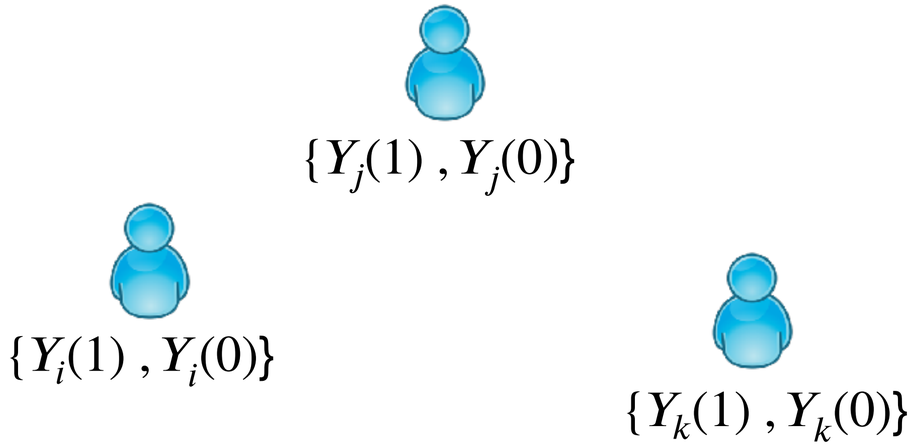
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Fundamental Problem of Causal Inference

How can we find the individual causal effect?

$$\tau_i = Y_i(1) - Y_i(0)$$

Option-1: Assume that $\{Y_i(1), Y_i(0)\}$ is the same for all individuals? (unit homogeneity)



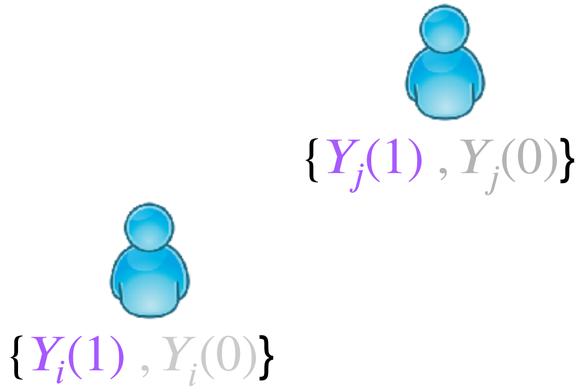
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Fundamental Problem of Causal Inference

How can we find the individual causal effect?

$$\tau_i = Y_i(1) - Y_i(0)$$

Option-1: Assume that $\{Y_i(1), Y_i(0)\}$ is the same for all individuals? (unit homogeneity), i.e., causal effect is the same across units



$$\tau_i = Y_i(1) - Y_k(0)$$
$$\tau_j = Y_j(1) - Y_k(0)$$
$$\tau_k = Y_i(1) - Y_k(0)$$

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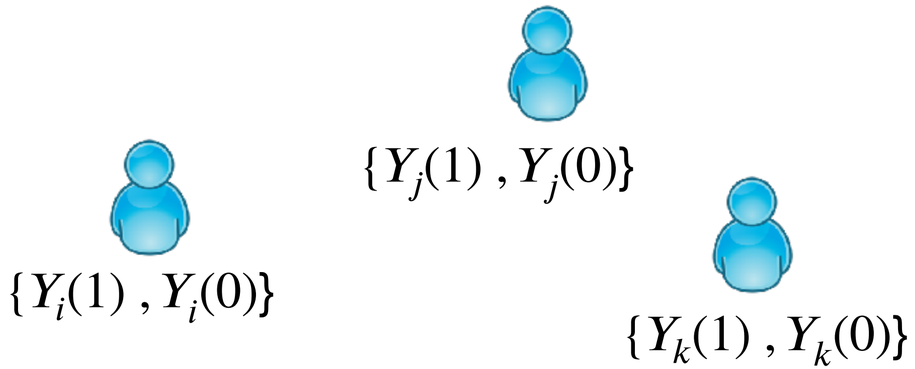
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Option-1: Assume that $\{Y_i(1), Y_i(0)\}$ is the same for all individuals? (unit homogeneity), i.e., causal effect is the same across units

Option-2: Assume that $\{Y_i(1), Y_i(0)\}$ is constant across time (though the effect may differ across units)



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Fundamental Problem of Causal Inference



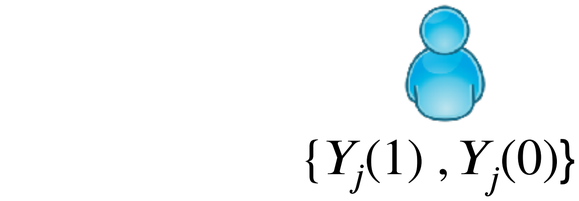
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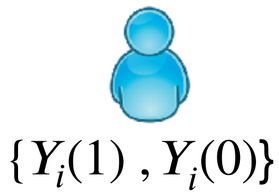
$$\tau_i = Y_i(1) - Y_i(0)$$

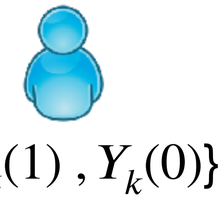
Option-1: Assume that $\{Y_i(1), Y_i(0)\}$ is the same for all individuals? (unit homogeneity) , i.e., causal effect is the same across units

Option-2: Assume that $\{Y_i(1), Y_i(0)\}$ is constant across time (though the effect may differ across units

$$\tau_i = Y_i(1) - Y_i(0)$$
$$\tau_j = Y_j(1) - Y_j(0)$$
$$\tau_k = Y_k(1) - Y_k(0)$$







Measured before treatment is given

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Fundamental Problem of Causal Inference



- These assumptions are strong, which are rarely true
  - Without any further information about the missing potential outcomes, we cannot estimate the individual causal effect

- So, what should we do?
  - Instead, we aim to estimate the average causal effect (a.k.a. average treatment effect
  - That is, we consider groups of individuals rather than one person and work with average treatment effect

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Average Treatment Effect (ATE)

- Average of over all individual causal effects

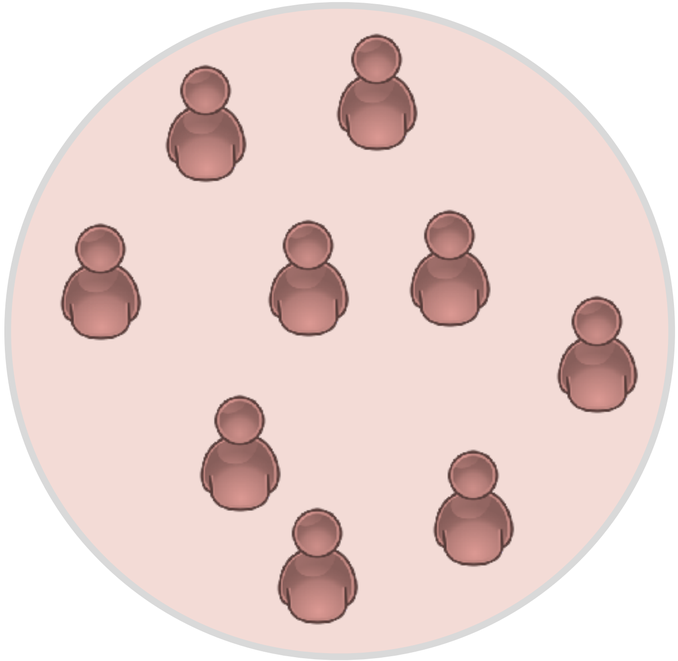
T_i	$Y_i(0)$	$Y_i(1)$
0	×	?
⋮	⋮	⋮
0	×	?
1	?	×
⋮	⋮	⋮
1	?	×

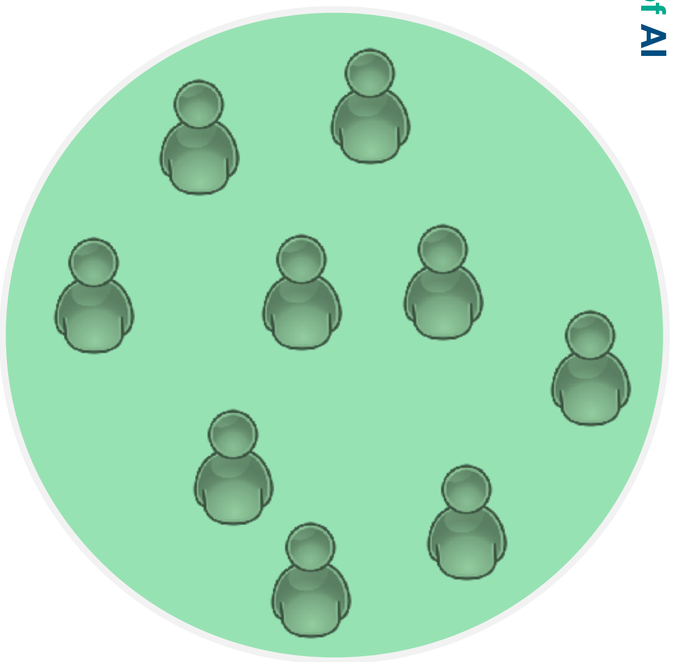
$$\mathbb{E}[\tau_i] = \mathbb{E}[Y_i(1) - Y_i(0)]$$

Can we estimate $\mathbb{E}[\tau_i]$ from data?

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Differences-in-Means Estimator for ATE





D: Difference in means (aka associational difference)

$$D = \mathbb{E}[Y_i(1) \mid T_i = 1] - \mathbb{E}[Y_j(0) \mid T_j = 0]$$

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Finding Average Causal Effects

Is ATE = D?

Is $\mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[Y_i(1) \mid T_i = 1] - \mathbb{E}[Y_j(0) \mid T_j = 0]$?

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Let's Work Through an Example



Foundations of AI

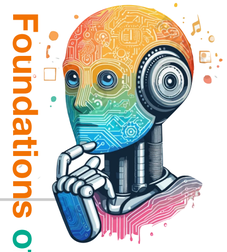
AI500

Suppose we could observe both potential outcomes for all individuals!

Person (i)	T_i	$Y_i(1)$	$Y_i(0)$	τ_i
1	1	0.6	0.3	0.3
2	0	0.2	0.6	-0.4
3	1	0.4	0.2	0.2
4	0	0.3	0.1	0.2
5	1	0.7	0.5	0.2
6	0	0.6	0.4	0.2
7	0	0.3	0.1	0.2

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Let's Work Through an Example



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4	0	0.3	0.1	0.2
5	1	0.7	0.5	0.2
6	0	0.6	0.4	0.2
7	0	0.3	0.1	0.2
		$\mathbb{E}[Y_i(1)] = ?$	$\mathbb{E}[Y_i(0)] = ?$	$\tau = \mathbb{E}[Y_i(1) - Y_i(0)] = ?$

Let's Work Through an Example



Suppose we could observe both potential outcomes for all individuals!

Person (i)	T_i	$Y_i(1)$	$Y_i(0)$	τ_i
1	1	0.6	0.3	0.3
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5	1	0.7	0.5	0.2
6	0	0.6	0.4	0.2
7	0	0.3	0.1	0.2
		$\mathbb{E}[Y_i(1)] = 0.44$	$\mathbb{E}[Y_i(0)] = 0.31$	$\tau = 0.13$

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But we don't know the $\tau_i's$!



Person (i)	T_i	$Y_i(1)$	$Y_i(0)$	τ_i
1	1	0.6	0.3	0.3
2	0	0.2	0.6	-0.4
3	1	0.4	0.2	0.2
4	0	0.3	0.1	0.2
5	1	0.7	0.5	0.2
6	0	0.6	0.4	0.2
7	0	0.3	0.1	0.2
		$\mathbb{E}[Y_i(1)T = 1] = ?$	$\mathbb{E}[Y_i(0) \mid T = 0] = ?$	$D = ?$

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But we don't know the $\tau_i's$!

Let's find the average of the observed outcomes in the two groups.

Person (i)	T_i	$Y_i(1)$	$Y_i(0)$	τ_i
1	1	0.6	0.3	0.3
2	0	0.2	0.6	-0.4
3	1	0.4	0.2	0.2
4	0	0.3	0.1	0.2
5	1	0.7	0.5	0.2
6	0	0.6	0.4	0.2
7	0	0.3	0.1	0.2
		$\mathbb{E}[Y_i(1)T = 1] = 0.57$	$\mathbb{E}[Y_i(0) \mid T = 0] = 0.3$	$D = 0.27$

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DIY

(age, gender, exercise)	Y_0: Sugar levels *had they received* medication A	Y_1: Sugar levels *had they received* medication B	Observed sugar levels
(45, F, 0)	6	5.5	6
(45, F, 1)	7	6.5	6.5
(55, M, 0)	7	6	7
(55, M, 1)	9	8	8
(65, F, 0)	8.5	8	8
(65,F, 1)	7.5	7	7.5
(75,M, 0)	10	9	9
(75,M, 1)	8	7	8

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DIY



Foundations of AI AI500

(age, gender, exercise)	Y_0: Sugar levels *had they received* medication A	Y_1: Sugar levels *had they received* medication B	Observed sugar levels
(45, F, 0)	6	5.5	6
(45, F, 1)	7	6.5	6.5
(55, M, 0)	7	6	7
(55, M, 1)	9	8	8
(65, F, 0)	8.5	8	8
(65,F, 1)	7.5	7	7.5
(75,M, 0)	10	9	9
(75,M, 1)	8	7	8

mean(sugar|medication B) –
mean(sugar|medicaton A) =
7.875 - 7.125 = 0.75

mean(sugar|*had they received* B) –
mean(sugar|*had they received* A) =
7.125 - 7.875 = -0.75

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Take-away

$$\mathbb{E}[\tau_i] = 0.13 \quad \neq \quad D = 0.27$$

- In general, D is NOT the same as ATE!
- Why? because association is not causation!

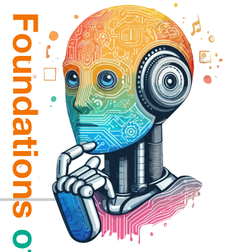


So what do we need for D to be equal to ATE?

Key Assumptions

- Ignorability: Potential outcomes are independent of the treatment assignment: $\{Y_i(1), Y_i(0)\} \perp T$
- Stable Unit Treatment Value Assumption (SUTVA) (a.k.a. no interference assumption)
  - Ex: vaccination, communication
- No different versions of treatment (a.k.a. Stability, consistency)
  - Violations may occur due to variable levels of treatment or technical errors

When is D = ATE?



1. $\mathbb{E}[Y_i(1) \mid T = 1] - \mathbb{E}[Y_i(0) \mid T = 0]$

2. $= \mathbb{E}[Y_i(1) \mid T = 1] - \mathbb{E}[Y_i(0) \mid T = 1] + \mathbb{E}[Y_i(0) \mid T = 1] - \mathbb{E}[Y_i(0) \mid T = 0]$

3. $= (\mathbb{E}[Y_i(1) \mid T = 1] - \mathbb{E}[Y_i(0) \mid T = 1]) + \mathbb{E}[Y_i(0) \mid T = 1] - \mathbb{E}[Y_i(0) \mid T = 0]$

$If\ \{Y_i(1), Y_i(0)\} \perp T\ for\ all\ i\ then$

4. $= (\mathbb{E}[Y_i(1) \mid T = 1] - \mathbb{E}[Y_i(0) \mid T = 1]) + \mathbb{E}[Y_i(0) \mid T = 1] - \mathbb{E}[Y_i(0) \mid T = 0]$

5. $= \mathbb{E}[Y_i(1) - Y_i(0)]$

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Causal Inference: Missing Data Problem



- Causal inference is the problem of estimating the counterfactual
  - Estimate missing data values using various methods
  - $Y_i(0)$ now becomes an estimated quantity, based on outcomes of other people who did not receive treatment

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Various Methodologies

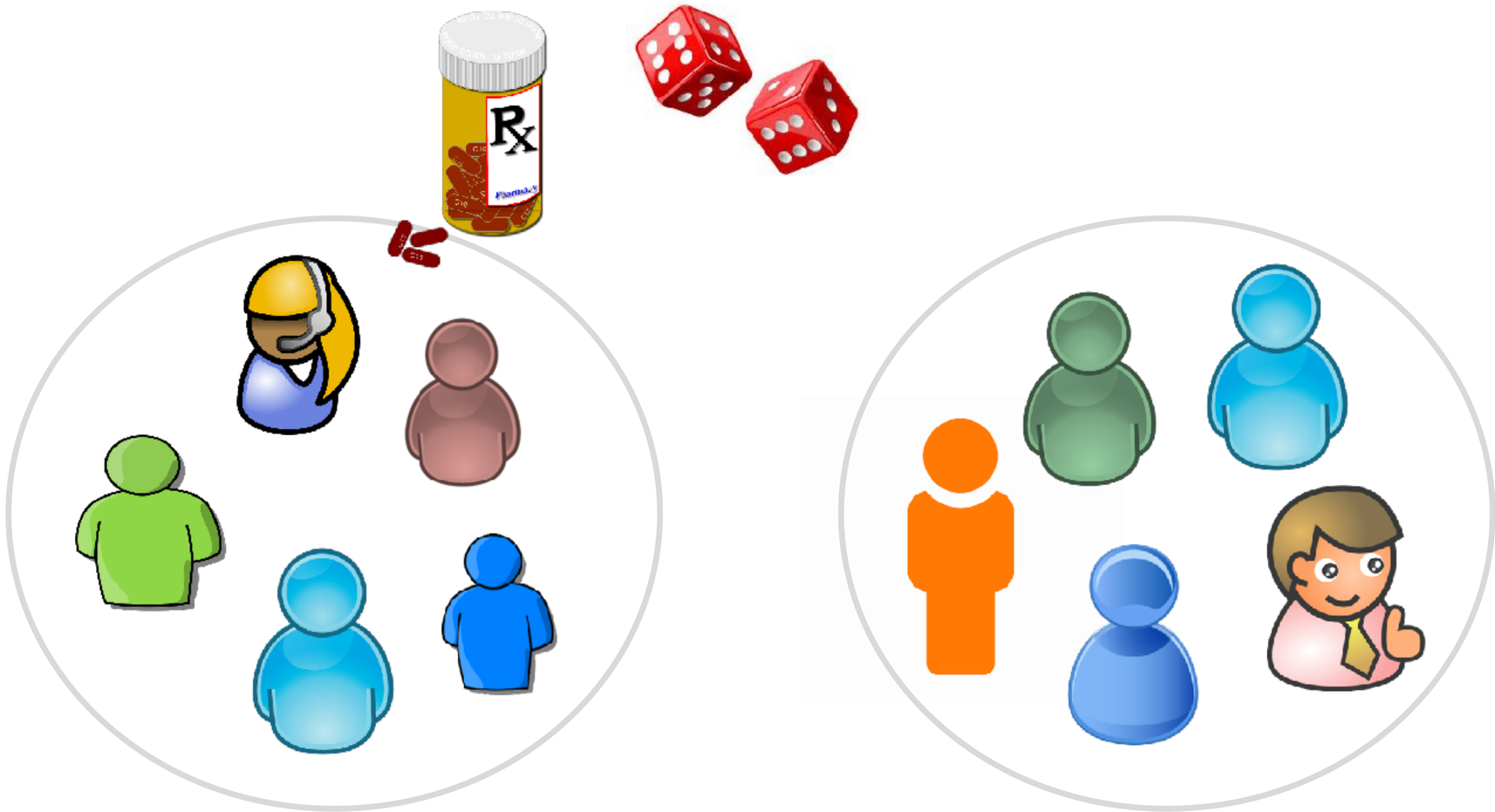


- Randomized Control Trials (RCTs)
  - Gold standard

- Quasi-Experimental Methods
  - Examples: cross-sectional approaches, before-after comparison, matching, difference-in-differences, regression discontinuity, and instrumental variables

Randomized Control Trials

One way to estimate the counterfactual; a gold standard



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